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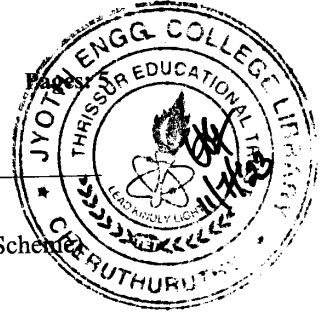
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Reg No.: \_\_\_\_\_

Name: \_\_\_\_\_

**APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY**

B.Tech Degree S6 (R, S) / S4 (PT) (R, S) Examination June 2023 (2019 Scheme)



**Course Code: EET304**

**Course Name: POWER SYSTEMS II**

Max. Marks: 100

Duration: 3 Hours

**PART A**

*Answer all questions, each carries 3 marks.*

Marks

- 1 Two generators rated 10MVA, 13.2kV and 15MVA, 13.2 kV are connected in parallel to a bus bar. They feed two motors of input 8MVA and 12MVA. The operating voltages of the motors are 12.5kV. Assuming base quantities as 50MVA and 13.8 kV, draw the reactance diagram. The percentage reactance of the generator is 15% and for motors is 20%. (3)
- 2 Discuss the effect of faults in a power system? Explain what do you mean by a symmetrical fault. (3)
- 3 Explain how are buses classified. (3)
- 4 A power system network has 100 buses. The Y-Bus matrix is 90 % sparse. Determine the number of transmission lines in the system. (3)
- 5 A 50 Hz synchronous generator is connected to an infinite bus through a line. The p.u. reactances of generator and the line are j0.3 p.u. and j0.3 p.u. respectively. The generator no load voltage is 1.2 p.u. and that of infinite bus is 1.0 p.u. The inertia constant of the generator is 3 MW-sec/MVA. Determine the frequency of natural oscillations if the generator is loaded 75% of its maximum power transfer capacity and small perturbation in power is given. (3)
- 6 Define synchronizing coefficient. For what value of synchronizing coefficient the system remains stable. (3)
- 7 Why should the system frequency be maintained constant? (3)
- 8 Differentiate between ACE in a single-area system and a two-area system. (3)
- 9 Discuss about thermal unit constraints in unit commitment problem? (3)
- 10 Explain how economic dispatch can be done when transmission losses are considered. (3)

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**PART B**

*Answer one full question from each module, each carries 14 marks.*

**Module I**

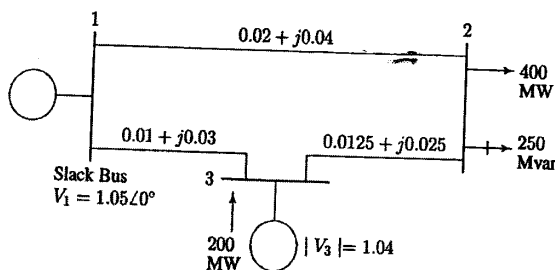
- 11 a) Derive the expression for fault current and draw the interconnection of sequence networks for line to ground fault on the terminals of an unloaded generator. (10)
- b) In a 3phase unsymmetrical system, the line currents flowing towards a delta connected load are  $I_a = 10 \angle -45^\circ$  and  $I_b = 10 \angle 90^\circ$ . Find the symmetrical components of line currents. (4)

**OR**

- 12 a) A synchronous generator and motor are rated 25MVA, 11 kV, having sub transient reactances of 15% are connected through transformers and a transmission line. The transformers are rated 25MVA, 11/66kV and 66/11kV with leakage reactance of 10% each. The line connecting them has a reactance of 10% on the base of 25MVA, 66kV. The motor is drawing 15MW at 0.8 p.f. leading and a terminal voltage of 10.6kV when a symmetrical three phase fault occurs at the motor terminals. Find the sub transient currents in the generator, motor and fault using the internal voltages of the machines. (10)
- b) Explain how circuit breaker ratings are chosen. (4)

**Module II**

- 13 a) For the system shown, obtain the power flow solution by Fast Decoupled Load flow. The voltage at bus 1 is adjusted to 1.05 p.u. Voltage magnitude at bus 3 is fixed at 1.04 p.u. with real power generation of 200MW. A load consisting of 400MW and 250MVAR is taken from bus 2. Line impedances are marked in per unit on a 100MVA base and line susceptances are neglected. (10)



- b) What is a Jacobian Matrix? How are the elements of the Jacobian matrix computed? (4)

**OR**

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- 14 a) The following is the system data for load flow solution. The line admittances are given in the following table. (10)

| Bus code | Admittance     |
|----------|----------------|
| 1-2      | $2-j8$         |
| 1-3      | $1-j4$         |
| 2-3      | $0.666-j2.664$ |
| 2-4      | $1-j4$         |
| 3-4      | $2-j8$         |

The schedule of active and reactive powers are:

| Bus Code | P   | Q   | V    | Remarks |
|----------|-----|-----|------|---------|
| 1        | -   | -   | 1.06 | Slack   |
| 2        | 0.5 | 0.2 | -    | PQ Bus  |
| 3        | 0.4 | 0.3 | -    | PQ Bus  |
| 4        | 0.3 | 0.1 | -    | PQ Bus  |

Determine the voltages at the end of first iteration. Take  $\alpha=1.6$ .

- b) Compare Gauss-Siedel and Newton Raphson load flow methods. (4)

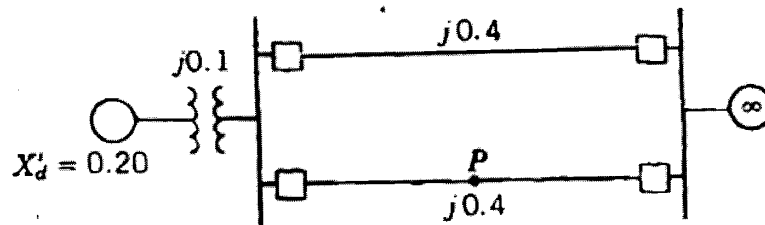
### Module III

- 15 a) State and explain equal area criterion. How do you apply equal area criterion to find the maximum additional load that can be suddenly added? (8)
- b) A 200 MVA, 11 KV 50Hz 4 pole turbo alternator has an inertia constant of 6MJ/MVA. Find i) stored energy in the rotor at synchronous speed. The machine is operating at a load of 120MW. When load increases to 160MW, ii) find the rotor acceleration, neglecting losses. (6)

OR

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- 16 a) The single-line diagram of the figure shows a power system, where power is fed through parallel transmission lines to a large metropolitan system considered as an infinite bus. The machine is delivering 1.0 per unit power and both the terminal voltage and infinite-bus voltage are both 1.0 per unit. The reactances of all components are shown on the diagram. a) Determine the power angle equation for the given system. (8)
- b) If a three phase fault occurs at point P, find the new power angle equation. Plot the two curves.



- b) What is critical clearing angle and time? Obtain expressions for the same. (6)

#### Module IV

- 17 a) Derive from fundamentals, the complete block diagram of load frequency control of an isolated power system. (10)
- b) A 125 MVA turbo-alternator on full load operates at 50 Hz. A load of 50 MW is suddenly reduced on the machine. The steam valves to the turbine commence to close after 0.5 s due to the time lag in the governor system. Assuming the inertia to be constant,  $H = 6$  kW/s per kVA of generator capacity, calculate the change in frequency that occurs in this time. (4)

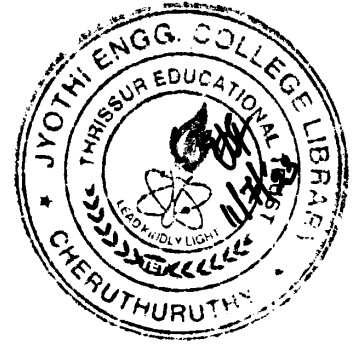
OR

- 18 a) Explain about proportional plus integral LFC with a block diagram. (10)
- b) A 500-MW generator has a speed regulation of 4%. If the frequency drops by 0.12 Hz with an unchanged reference, determine the increase in turbine power. (4)
- And also find by how much the reference power setting should be changed if the turbine power remains unchanged.

#### Module V

- 19 a) Determine the saving in fuel cost in Rs./hr for the economic distribution of a (10)

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total load of 225 MW between the two units with IFCs:

$$\frac{dF_1}{dP_1} = 0.075P_1 + 15$$

$$\frac{dF_2}{dP_2} = 0.085P_2 + 12.$$

Compare with equal distribution of the same total load.

- b) What are B-coefficients? Write the expression for transmission loss in terms of B - coefficients when there are three generating stations. (4)

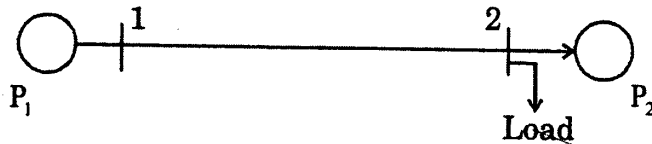
OR

- 20 a) A two bus system is shown in figure. If a load of 120MW is transmitted from plant 1 to a load a loss of 16.425MW is incurred. Determine the optimal scheduling of plants and the load demand if the cost of received power is Rs. 26/MWh. (10)

. The incremental production costs of the plants are

$$\frac{\partial F_1}{\partial P_1} = 0.015P_1^2 + 18P_1 + 20$$

$$\frac{\partial F_2}{\partial P_2} = 0.03P_1^2 + 33P_2 + 40$$



- b) What do you mean by spinning reserve? Why the spinning reserve must be maintained. (4)

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